1 What are different sand moulding machines / equipments?

A sand moulding machine may be defined as a device used for compacting or moulding sand. These machines can be classified as

(i) Squeezers

A squeezer machine uses air pressure as a measure of compacting the sand. The force of application of a pneumatically operated squeezer can be calculated by formula

$$F = P\frac{\pi D^2}{4} - W$$

where,
F → total force of compression of machine.
P → pneumatic pressure cylinder.
D → dia of squeeze cylinder
W → Weight of the patter, flask and other accessories.

ii) Jolt machine

It is also known as a jar machine or bumper. The work table with pattern flask and sand is raised by a pneumatically operated piston, when the piston, and hence the table has been raised to a certain height the air below the piston is suddenly released. The fall of the table against the base of machine takes place due to force of gravity. It results in packing of the moulding sand due to work done by kinetic energy of falling sand. The power of jolting produced due to the sudden fall of the table is given by.

Power of jolting = $\frac{wv}{A}$

$w \rightarrow$ weight of sand

$v \rightarrow$ velocity jst before jolt (in m)

$A \rightarrow$ jolted area (m^2)

Jolt Sqeezer

(iii) Jolt Squeezer machine utilises a combination of jolting and squeezing to pack moulding sand and making the mould. A jolt machine produces a good compaction around the pattern of upto 5 cm thickness. Squeezers produce good compaction at the top of a mould. Jolt squeezing machine eliminates the defect of jolt machine by & squeezer machine and utillizes the beneficial effects of the two. In fact, this is the reason by jolt squeezer moulding machines are more popular than jolt machines or squeezing machines.

(iv) Sand Slingers

Sand slingers are moulding machines used for filling and compacting sand in the moulds at the same time. In these machines ramming is obtained by impacting of sand.

(v) Blower.

A mould blower is pneumatic form of slinging that is more commonly used to making cores than moulds. It had estect itself as the principal means of rapid production of small and medium sized cores. In core blowing machine the core box is simultaneously filled and rammed with and carrying a steam of air

Explain chvorinovs rule and cains rule for solidification time.

Chvorinovs rule

Solidification time or freezing time,

$$t = C\left(\frac{V}{S.A}\right)^2$$

where V → Volume of casting

S.A → Surface area of casting

C → Constant of proper timing that depends upon composition / properties of cast metal (including its latent heat) pouring temp. mould material etc...

Since the metal in the riser must be the last to solidify to achieve directional solidification.

$$\left(\frac{V}{S.A}\right)_{riser} > \left(\frac{V}{S.A}\right)_{casting}$$

Since V and SA for the casting are known. $\left(\frac{V}{SA}\right)$ riser can be determined. Assuming height to dia ratio for the cylinder riser, the riser size can be determined.

Caines Formulae

Caines method of determining riser size is based on an experimentally determined hyperbolic relationship b/w relative freezing time and volumes of the casting and the riser.

→ Relative freezing time or freezing ratio is determined as

$$X = \frac{(S.A/V)_{casting}}{(S.A/V)_{riser}}$$

volume ratio, y is given as:

$$y = \frac{\text{volume of riser}}{\text{volume of casting}}$$

Then caines formula is given as

$$x = \frac{a}{y-b} + c.$$

where
a = Freezing characteristic constant
b = Liquid-solid solidification contraction
c = Relative freezing rate of riser and casting

What are product design consideration for defect free casting?

There are several consideration for defect free casting some are;

i) Compensate the shrinkage of the solidified molten metal by making patterns of slightly oversize.

i) In sand casting it is more economical and accurate if the parting line is on a flat plane. Countoured parting lines are not economical. Further some degree of taper or drift is recommended to provide to the pattern for easy removal.

ii) In sand casting it is recomended to attach the raiser near to the heavier section. The thinnest section are farthest and the raiser and solidify first.

i) sharp corners in a castong design cause uneven cooling and lead to formation of hot spots in final cast structure.

moreover, sharp corners in a casting structure acts as a stress raiser. Rounding the corner decreases the severity of the hotspot and lessens the stress concentration.

v) Abrupt changes in section should be avoided. fillets and tapers are preferabble to sharp steps.

vi) The Interior walls and sections are recommended to be 20% less thinner than the outside member to reduce the thermal and residual stressess. and metallurgical changes.

(vii) when a hole is placed in a highly stressed section. add extra material around the hole as reinforcements.

(viii) To minimise the residual stress in the gear, pulley or wheel casting, a balance b/w the section size of rim, spokes and hub is maintained.

(ix) An odd number of curved wheel spokes reduce cast-in-residual stresses.

4. Explain the process of wax casting.

Low-wax process, also called cireperdure method of metal casting in which a molten metal is poured into a mould that has been created by means of wax model. Once the mould is made the wax model is melted and drained away. A hollow core can be effected by introduction of heat proce· core that prevents molten metal from totally filling the mould.